

LIAM TAYLOR

(813)644-1418 | liammtaylor@usf.edu | Tampa, FL | <https://www.linkedin.com/in/liamtaylor99>

Education

Bachelor of Science in Electrical Engineering, Concentration in Power Systems

Jan 2026-Present

University of South Florida, Tampa, FL

Bachelor of Science in Physical Sciences

Dec 2024

University of South Florida, Tampa, FL

Relevant coursework : Electrical Systems I & II, Digital Circuits, Electronic Materials, Engineering Analysis, Material and Energy Balances, Transport phenomena, Thermodynamics, Organic and Physical Chemistry, Modern Physics

Research Experience

Undergraduate Research Assistant, TROPICS Lab, Dept. of Electrical Engineering, USF

Sept 2026 – Present

- Working under PhD mentors Mrinmoy Turja and Mitchell Harrah on Diffuse Correlation Spectroscopy (DCS) instrumentation — avalanche photon-counting paired with FPGA/correlator data acquisition for noninvasive deep-tissue blood-flow measurement — supporting a first-of-its-kind DCS sonogram for placental blood-vessel health before full fetal development.

Work Experience

Assembly Technician 2, Lithionics, Clearwater, FL

Jan 2025 – Jul 2026

- Performed First Article Inspections on 400A-rated UL-certified Battery Management Systems and 12V/24V/51V/102V single-module lithium-ion batteries, verifying internal data-monitoring components and physical build integrity.
- Conducted discharge testing and final-approval inspections on multi-module battery systems, maintaining in-house data records against customer specifications and using findings to drive process-checklist improvements.
- Collaborated with engineering to redesign a power-connection voltage/polarity testing tool now standard in the production process, reducing defect rates and improving technician safety during module testing.

Assistant Foreman, TE Taylor Co., Tampa, FL

Jun 2018-Aug 2024

- Direct and coordinate multiple company leaders and crew members in commercial development of ADA-compliant concrete sidewalks, ramps, and drainage valleys
- Evaluate site conditions and project plans with head foreman to optimize workflow, reducing overtime and increasing completion rates

Personal Projects

Court Commander, Personal Project — Vanilla JavaScript, HTML5 Canvas, SheetJS

- Built a browser-based volleyball coaching application handling live scoring, automatic rotation logic, dual-team lineup management, and a 6-rotation planner with stat-tracking and XLSX export via SheetJS — used as the primary in-game tool for a competitive high school program.

IEEE EXO — Project Lead, Robotic Gait-Assist Leg System

- Leading a cross-organizational robotic exoskeleton project spanning IEEE EXO, a biomedical engineering club, and a campus research lab, coordinating six subsystem teams against a two-semester timeline and \$5,000 budget; authored the project's governance and requirements framework to drive accountable, independent parallel workstreams.

Mime, IEEE EXO Flagship Project — Gesture-Controlled Robotic Hand

- Developed a dual-mode (MediaPipe webcam vision and flex-sensor glove) gesture control system for a 5-finger tendon-driven robotic hand on an ESP32/PCA9685 platform, with calibration tooling and a debounced real-time gesture interface.

Copper Electroplating, Material Science / Electrochemistry Project

- Electroplated a copper coating onto an organic specimen using a conductive graphite-painted cathode in a copper sulfate electrolytic cell with a copper-wire anode and 6V DC source, depositing over 10g of copper in 6 hours; applied Faraday's laws of electrolysis to relate deposited mass to charge and current.

Technical Skills

Technologies: C, Python (pandas, numpy, astropy), MATLAB/Simulink, HTML, Java, JavaScript, CSS, Fusion 360, KiCad

Certification : SACHe Certificates (Process Safety Hazards, Process Safety Intro, Hazard Recognition), CPR

Software: Claude, Claude Code, VS Code, GitHub Copilot, Bootstrap, Microsoft Office (Excel, Word, PowerPoint)